CHAPTER 17 SUMMARY

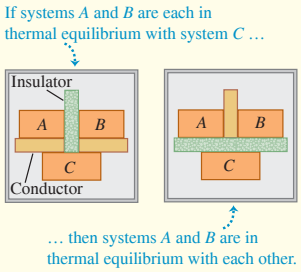
Temperature and temperature scales: Two objects in thermal equilibrium must have the same temperature. A conducting material between two objects permits them to interact and come to thermal equilibrium; an insulating material impedes this interaction.

The Celsius temperature scale is based on the freezing (0°C) and boiling (100°C) temperatures of water.

The Kelvin scale has its zero at the extrapolated zero-pressure temperature for a gas thermometer, $-273.15°\text{C} = 0\ \text{K}$. In the gas-thermometer scale, the ratio of two temperatures T_1 and T_2 is defined to be equal to the ratio of the two corresponding gas-thermometer pressures p_1 and p_2.

$$T_K = T_C + 273.15 \quad (17.1)$$

$$\frac{T_2}{T_1} = \frac{p_2}{p_1} \quad (17.2)$$



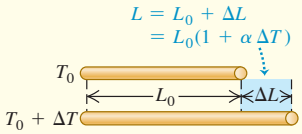
Thermal expansion and thermal stress: A temperature change ΔT causes a change in any linear dimension L_0 of a solid object. The change ΔL is approximately proportional to L_0 and ΔT. Similarly, a temperature change causes a change ΔV in the volume V_0 of any solid or liquid; ΔV is approximately proportional to V_0 and ΔT. The quantities α and β are the coefficients of linear expansion and volume expansion, respectively. For solids, $\beta = 3\alpha$. (See Examples 17.2 and 17.3.)

When a material is cooled or heated and held so it cannot contract or expand, it is under a tensile stress F/A. (See Example 17.4.)

$$\Delta L = \alpha L_0\, \Delta T \quad (17.4)$$

$$\Delta V = \beta V_0\, \Delta T \quad (17.6)$$

$$\frac{F}{A} = -Y\alpha\, \Delta T \quad (17.10)$$



Heat, phase changes, and calorimetry: Heat is energy in transit from one object to another as a result of a temperature difference. Equations (17.11) and (17.16) give the quantity of heat Q required to cause a temperature change ΔT in a quantity of material with mass m and specific heat c (alternatively, with number of moles n and molar heat capacity $C = Mc$, where M is the molar mass and $m = nM$). When heat is added to an object, Q is positive; when it is removed, Q is negative. (See Examples 17.5 and 17.6.)

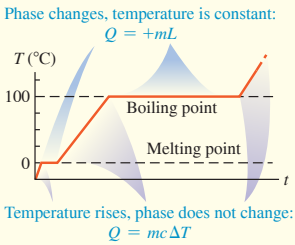
To change a mass m of a material to a different phase at the same temperature (such as liquid to vapor), a quantity of heat given by Eq. (17.18) must be added or subtracted. Here L is the heat of fusion, vaporization, or sublimation.

In an isolated system whose parts interact by heat exchange, the algebraic sum of the Q's for all parts of the system must be zero. (See Examples 17.7–17.10.)

$$Q = mc\, \Delta T \quad (17.11)$$

$$Q = nC\, \Delta T \quad (17.16)$$

$$Q = \pm mL \quad (17.18)$$



Conduction, convection, and radiation: Conduction is the transfer of heat within materials without bulk motion of the materials. The heat current H depends on the area A through which the heat flows, the length L of the heat-flow path, the temperature difference $(T_H - T_C)$, and the thermal conductivity k of the material. (See Examples 17.11–17.13.)

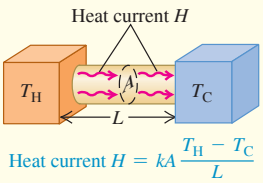
Convection is a complex heat-transfer process that involves mass motion from one region to another.

Radiation is energy transfer through electromagnetic radiation. The radiation heat current H depends on the surface area A, the emissivity e of the surface (a pure number between 0 and 1), and the Kelvin temperature T. Here σ is the Stefan–Boltzmann constant. The *net* radiation heat current H_{net} from an object at temperature T to its surroundings at temperature T_s depends on both T and T_s. (See Examples 17.14 and 17.15.)

$$H = \frac{dQ}{dt} = kA\frac{T_H - T_C}{L} \quad (17.19)$$

$$H = Ae\sigma T^4 \quad (17.23)$$

$$H_{net} = Ae\sigma(T^4 - T_s^4) \quad (17.24)$$



Chapter 17 Media Assets

For assigned homework and other learning materials, go to Mastering Physics.

GUIDED PRACTICE

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review EXAMPLES 17.2, 17.3, and 17.4 (Section 17.3) before attempting these problems.

VP17.4.1 A metal rod is 0.500 m in length at a temperature of 15.0°C. When you raise its temperature to 37.0°C, its length increases by 0.220 mm. (a) What is the coefficient of linear expansion of the metal? (b) If a second rod of the same metal has length 0.300 m at 25.0°C, how will its length change if the temperature drops to −20.0°C?

VP17.4.2 A copper mug that can hold 250 cm^3 of liquid is filled to the brim with ethanol at 20.0°C. If you lower the temperature of the mug and ethanol to −50.0°C, what is the maximum additional volume of ethanol you can add to the mug without spilling any? (See Table 17.2. Ethanol remains a liquid at temperatures down to −114°C.)

VP17.4.3 A cylindrical brass rod is 10.0 cm in length and 0.500 cm in radius at 25.0°C. How much force do you have to apply to each end of the rod to maintain its length when the temperature is decreased to 13.0°C? Are the required forces tensile or compressive? (See Table 17.1. Brass has Young's modulus 9.0×10^{10} Pa.)

VP17.4.4 A rod made of metal A is attached end to end to another rod made of metal B, making a combined rod of overall length L. The coefficients of linear expansion of metals A and B are α_A and α_B, respectively, and $\alpha_B > \alpha_A$. When the temperature of the combined rod is increased by ΔT, the overall length increases by ΔL. What was the initial length of the rod of metal A?

Be sure to review EXAMPLES 17.7, 17.8, and 17.9 (Section 17.5) before attempting these problems.

VP17.9.1 You place a piece of aluminum at 250.0°C in 5.00 kg of liquid water at 20.0°C. None of the water boils, and the final temperature of the water and aluminum is 22.0°C. What is the mass of the piece of aluminum? Assume no heat is exchanged with the container that holds the water. (See Table 17.3.)

VP17.9.2 You place an ice cube of mass 7.50×10^{-3} kg and temperature 0.00°C on top of a copper cube of mass 0.460 kg. All of the ice melts, and the final equilibrium temperature of the two substances is 0.00°C. What was the initial temperature of the copper cube? Assume no heat is exchanged with the surroundings. (See Tables 17.3 and 17.4.)

VP17.9.3 You have 1.60 kg of liquid ethanol at 28.0°C that you wish to cool. What mass of ice at initial temperature −5.00°C should you add to the ethanol so that all of the ice melts and the resulting ethanol–water mixture has temperature 10.0°C? Assume no heat is exchanged with the container that holds the ethanol. (See Tables 17.3 and 17.4.)

VP17.19.4 You put a silver ingot of mass 1.25 kg and initial temperature 315°C in contact with 0.250 kg of ice at initial temperature −8.00°C. Assume no heat is exchanged with the surroundings. (a) What is the final equilibrium temperature? (b) What fraction of the ice melts? (See Tables 17.3 and 17.4.)

Be sure to review EXAMPLES 17.11, 17.12, 17.13, 17.14, and 17.15 (Section 17.6) before attempting these problems.

VP17.15.1 A square pane of glass 0.500 m on a side is 6.00 mm thick. When the temperatures on the two sides of the glass are 25.0°C and −10.0°C, the heat current due to conduction through the glass is 1.10×10^3 W. (a) What is the thermal conductivity of the glass? (b) If the thickness of the glass is increased to 9.00 mm, what will be the heat current?

VP17.15.2 A brass rod and a lead rod, each 0.250 m long and each with cross-sectional area 2.00×10^{-4} m^2, are joined end to end to make a composite rod of overall length 0.500 m. The free end of the brass rod is maintained at a high temperature, and the free end of the lead rod is maintained at a low temperature. The temperature at the junction of the two rods is 185°C, and the heat current due to conduction through the composite rod is 6.00 W. What are the temperatures of (a) the free end of the brass rod and (b) the free end of the lead rod? (See Table 17.5.)

VP17.15.3 The emissivity of the surface of a star is approximately 1. The star Sirius A emits electromagnetic radiation at a rate of 9.7×10^{27} W and has a surface temperature of 9940 K. What is the radius of Sirius in meters and as a multiple of the sun's radius (6.96×10^8 m)?

VP17.15.4 A building in the desert is made of concrete blocks (emissivity 0.91) and has an exposed surface area of 525 m^2. If the building is maintained at 20.0°C but the temperature on a hot desert night is 35.0°C, what is the net rate at which the building absorbs energy by radiation?

BRIDGING PROBLEM Steady-State Heat Flow: Radiation and Conduction

One end of a solid cylindrical copper rod 0.200 m long and 0.0250 m in radius is inserted into a large block of solid hydrogen at its melting temperature, 13.84 K. The other end is blackened and exposed to thermal radiation from surrounding walls at 500.0 K. (Some telescopes in space employ a similar setup. A solid refrigerant keeps the telescope very cold—required for proper operation—even though it is exposed to direct sunlight.) The sides of the rod are insulated, so no energy is lost or gained except at the ends of the rod. (a) When equilibrium is reached, what is the temperature of the blackened end? The thermal conductivity of copper at temperatures near 20 K is 1670 W/m · K. (b) At what rate (in kg/h) does the solid hydrogen melt?

SOLUTION GUIDE

IDENTIFY and SET UP

1. Draw a sketch of the situation, showing all relevant dimensions.
2. List the known and unknown quantities, and identify the target variables.
3. In order for the rod to be in equilibrium, how must the radiation heat current from the walls into the blackened end of the rod compare to the conduction heat current from this end to the other end and into the solid hydrogen? Use your answers to select the appropriate equations for part (a).
4. How does the heat current from the rod into the hydrogen determine the rate at which the hydrogen melts? (*Hint:* See Table 17.4.) Use your answer to select the appropriate equations for part (b).

EXECUTE

5. Solve for the temperature of the blackened end of the rod. (*Hint:* Since copper is an excellent conductor of heat at low temperature, you can assume that the temperature of the blackened end is only slightly higher than 13.84 K.)
6. Use your result from step 5 to find the rate at which the hydrogen melts.

EVALUATE

7. Is your result from step 5 consistent with the hint in that step?
8. How would your results from steps 5 and 6 be affected if the rod had twice the radius?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q17.1 Explain why it would not make sense to use a full-size glass thermometer to measure the temperature of a thimbleful of hot water.

Q17.2 If you heat the air inside a rigid, sealed container until its Kelvin temperature doubles, the air pressure in the container will also double. Is the same true if you double the Celsius temperature of the air in the container? Explain.

Q17.3 Many automobile engines have cast-iron cylinders and aluminum pistons. What kinds of problems could occur if the engine gets too hot? (The coefficient of volume expansion of cast iron is approximately the same as that of steel.)

Q17.4 Why do frozen water pipes burst? Would a mercury thermometer break if the temperature went below the freezing temperature of mercury? Why or why not?

Q17.5 Two objects made of the same material have the same external dimensions and appearance, but one is solid and the other is hollow. When their temperature is increased, is the overall volume expansion the same or different? Why?

Q17.6 Why is it sometimes possible to loosen caps on screw-top bottles by dipping the capped bottle briefly into hot water?

Q17.7 The inside of an oven is at a temperature of 200°C. You can put your hand in the oven without injury as long as you don't touch anything. But since the air inside the oven is also at 200°C, why isn't your hand burned just the same?

Q17.8 A newspaper article about the weather states "the temperature of an object measures how much heat the object contains." Is this description correct? Why or why not?

Q17.9 A student asserts that a suitable unit for specific heat is $1\ \mathrm{m^2/s^2 \cdot {}^\circ C}$. Is she correct? Why or why not?

Q17.10 In some household air conditioners used in dry climates, air is cooled by blowing it through a water-soaked filter, evaporating some of the water. How does this cool the air? Would such a system work well in a high-humidity climate? Why or why not?

Q17.11 The units of specific heat c are $\mathrm{J/kg \cdot K}$, but the units of heat of fusion L_f or heat of vaporization L_v are $\mathrm{J/kg}$. Why do the units of L_f and L_v not include a factor of $(\mathrm{K})^{-1}$ to account for a temperature change?

Q17.12 Why is a hot, humid day in the tropics generally more uncomfortable for human beings than a hot, dry day in the desert?

Q17.13 A piece of aluminum foil used to wrap a potato for baking in a hot oven can usually be handled safely within a few seconds after the potato is removed from the oven. The same is not true of the potato, however! Give two reasons for this difference.

Q17.14 Desert travelers sometimes keep water in a canvas bag. Some water seeps through the bag and evaporates. How does this cool the water inside the bag?

Q17.15 When you first step out of the shower, you feel cold. But as soon as you are dry you feel warmer, even though the room temperature does not change. Why?

Q17.16 The climate of regions adjacent to large bodies of water (like the Pacific and Atlantic coasts) usually features a narrower range of temperature than that of regions far from them (like the prairies). Why?

Q17.17 When water is placed in ice-cube trays in a freezer, why doesn't the water freeze all at once when the temperature has reached 0°C? In fact, the water freezes first in a layer adjacent to the sides of the tray. Why?

Q17.18 Before giving you an injection, a physician swabs your arm with isopropyl alcohol at room temperature. Why does this make your arm feel cold? (*Hint:* The reason is *not* the fear of the injection! The boiling point of isopropyl alcohol is 82.4°C.)

Q17.19 A cold block of metal feels colder than a block of wood at the same temperature. Why? A *hot* block of metal feels hotter than a block of wood at the same temperature. Again, why? At what temperature, if any, do the two blocks feel equally hot or cold? What temperature is this?

Q17.20 A person pours a cup of hot coffee, intending to drink it five minutes later. To keep the coffee as hot as possible, should she put cream in it now or wait until just before she drinks it? Explain.

Q17.21 When a freshly baked apple pie has just been removed from the oven, the crust and filling are both at the same temperature. Yet if you sample the pie, the filling will burn your tongue but the crust will not. Why is there a difference? (*Hint:* The filling is moist while the crust is dry.)

Q17.22 Old-time kitchen lore suggests that things cook better (evenly and without burning) in heavy cast-iron pots. What desirable characteristics do such pots have?

Q17.23 In coastal regions in the winter, the temperature over the land is generally colder than the temperature over the nearby ocean; in the summer, the reverse is usually true. Explain. (*Hint:* The specific heat of soil is only 0.2–0.8 times as great as that of water.)

Q17.24 It is well known that a potato bakes faster if a large nail is stuck through it. Why? Does an aluminum nail work better than a steel one? Why or why not? (*Note:* Don't try this in a microwave oven!) There is also a gadget on the market to hasten the roasting of meat; it comprises a hollow metal tube containing a wick and some water. This is claimed to work much better than a solid metal rod. How does it work?

Q17.25 Glider pilots know that thermal updrafts are likely to occur in the vicinity of freshly plowed fields. Why?

Q17.26 Some folks claim that ice cubes freeze faster if the trays are filled with hot water, because hot water cools off faster than cold water. What do you think?

Q17.27 We're lucky that the earth isn't in thermal equilibrium with the sun (which has a surface temperature of 5800 K). But why isn't it?

Q17.28 When energy shortages occur, magazine articles sometimes urge us to keep our homes at a constant temperature day and night to conserve fuel. They argue that when we turn down the heat at night, the walls, ceilings, and other areas cool off and must be reheated in the morning. So if we keep the temperature constant, these parts of the house will not cool off and will not have to be reheated. Does this argument make sense? Would we really save energy by following this advice?

EXERCISES

Section 17.2 Thermometers and Temperature Scales

17.1 • With the temperature of −67.7°C recorded on February 6, 1933, Oymyakon in Sakha (Yakutia), Republic of Russia, is considered to be the coldest permanently inhabited place on earth. However, summer months in Oymyakon can be very warm, even hot. On July 28, 2010, the temperature reached 34.6°C. (a) Express both temperatures on the Kelvin scale. (b) Calculate the variation of temperatures both in degrees Celsius and in kelvins.

17.2 • **BIO** **Temperatures in Biomedicine.** (a) **Normal body temperature.** The average normal body temperature measured in the mouth is 310 K. What would a Celsius thermometer read for this temperature? (b) **Elevated body temperature.** During vigorous exercise, the body's temperature can go as high as 40°C. What would a Kelvin thermometer read for this temperature? (c) **Temperature difference in the body.** The surface

temperature of the body is normally about 7°C lower than the internal temperature. Express this temperature difference in kelvins. (d) **Blood storage.** Blood stored at 4°C lasts safely for about 3 weeks, whereas blood stored at −160°C lasts for 5 years. Express both temperatures on the Kelvin scale. (e) **Heat stroke.** If the body's temperature is above 41°C for a prolonged period, heat stroke can result. Express this temperature on Kelvin scale.

17.3 • Convert the following Kelvin temperatures to the Celsius scale: (a) the midday temperature at the surface of the moon (400 K); (b) the temperature at the tops of the clouds in the atmosphere of Saturn (95 K); (c) the temperature at the center of the sun $(1.55 \times 10^7\ \text{K})$.

17.4 • The pressure of a gas at the triple point of water is 1.35 atm. If its volume remains unchanged, what will its pressure be at the temperature at which CO_2 solidifies?

17.5 •• **A Constant-Volume Gas Thermometer.** An experimenter using a gas thermometer found the pressure at the triple point of water (0.01°C) to be 4.80×10^4 Pa and the pressure at the normal boiling point (100°C) to be 6.50×10^4 Pa. (a) Assuming that the pressure varies linearly with temperature, use these two data points to find the Celsius temperature at which the gas pressure would be zero (that is, find the Celsius temperature of absolute zero). (b) Does the gas in this thermometer obey Eq. (17.2) precisely? If that equation were precisely obeyed and the pressure at 100°C were 6.50×10^4 Pa, what pressure would the experimenter have measured at 0.01°C? (As we'll learn in Section 18.1, Eq. (17.2) is accurate only for gases at very low density.)

17.6 •• A constant-volume gas thermometer registers an absolute pressure corresponding to 325 mm of mercury when in contact with water at the triple point. What pressure does it read when in contact with water at the normal boiling point?

The Fahrenheit scale. In this temperature scale, still used in the United States, the freezing temperature of water is 32°F and the boiling temperature is 212°F, both at standard atmospheric pressure (see Appendix C). The formula to convert a temperature T_F in degrees Fahrenheit into a temperature T_C in degrees Celsius is $T_C = \frac{5}{9}(T_F - 32)$ and the formula to convert a temperature T_C given in degrees Celsius into a temperature T_F given in degrees Fahrenheit is $T_F = \frac{9}{5}T_C + 32$. Consequently, for temperature differences we have $\Delta T_C = \frac{5}{9}\Delta T_F$ and $\Delta T_F = \frac{9}{5}\Delta T_C$.

17.7 • Convert the following record-setting Celsius temperatures to Fahrenheit: (a) −89.2°C, the lowest temperature ever recorded (using ground-based equipment) in the world (July 21, 1983, Vostok Research Station, Antarctica); (b) 56.7°C, the highest air temperature ever recorded in the world (July 10, 1913, Death Valley, California, USA); (c) 30.7°C average annual temperature, Mecca, Saudi Arabia, the hottest city in the world.

17.8 • (a) On January 22, 1943, the temperature in Spearfish, South Dakota, USA, rose from −4.0°F to 45.0°F in just 2 minutes, the quickest temperature change ever recorded. What was the temperature change in degrees Celsius? (b) The temperature in Browning, Montana, was 44.0°F on January 23, 1916. The next day the temperature plummeted to −56.0°F, the largest temperature change in 24 hours ever recorded. What was the temperature change in degrees Celsius?

17.9 • Derive an equation that gives T_K as a function of T_F to the nearest hundredth of a degree. Solve the equation and thereby obtain an equation for T_F as a function of T_K.

17.10 • (a) Calculate the one temperature at which Fahrenheit and Celsius thermometers agree with each other. (b) Calculate the one temperature at which Fahrenheit and Kelvin thermometers agree with each other.

Section 17.3 Thermal Expansion

17.11 • The Humber Bridge in England has the world's longest single span, 1410 m. Calculate the change in length of the steel deck of the span when the temperature increases from −6.0°C to 17.5°C.

17.12 • One of the tallest buildings in the world is the Taipei 101 in Taiwan, at a height of 508 meters. Assume that this height was measured on a cool spring day when the temperature was 15.5°C. You could use the building as a sort of giant thermometer on a hot summer day by carefully measuring its height. Suppose you do this and discover that the Taipei 101 is 14.35 cm taller than its official height. What is the temperature, assuming that the building is in thermal equilibrium with the air and that its entire frame is made of steel?

17.13 • A Swiss 1-franc coin has a diameter of 23.200 mm at 20.0°C. The coin is made of a copper-nickel alloy (75% copper and 25% nickel) for which the coefficient of linear expansion is $16.3 \times 10^{-6}\ \text{K}^{-1}$. What would its diameter be on a hot day in Dubai (45°C)? On a cold night in the mountains of Greenland (−53°C)?

17.14 • **Ensuring a Tight Fit.** Aluminum rivets used in airplane construction are made slightly larger than the rivet holes and cooled by "dry ice" (solid CO_2) before being driven. If the diameter of a hole is 4.500 mm, what should be the diameter of a rivet at 23.0°C if its diameter is to equal that of the hole when the rivet is cooled to −78.0°C, the temperature of dry ice? Assume that the expansion coefficient remains constant at the value given in Table 17.1.

17.15 •• A copper cylinder is initially at 20.0°C. At what temperature will its volume be 0.150% larger than it is at 20.0°C?

17.16 •• A geodesic dome constructed with an aluminum framework is a nearly perfect hemisphere; its diameter measures 54.0 m on a winter day at a temperature of −12°C. How much more interior space does the dome have in the summer, when the temperature is 35°C?

17.17 •• A glass flask whose volume is 1000.00 cm^3 at 0.0°C is completely filled with mercury at this temperature. When flask and mercury are warmed to 55.0°C, 8.95 cm^3 of mercury overflow. If the coefficient of volume expansion of mercury is $18.0 \times 10^{-5}\ \text{K}^{-1}$, compute the coefficient of volume expansion of the glass.

17.18 •• A steel tank is completely filled with 2.90 m^3 of ethanol when both the tank and the ethanol are at 33.0°C. When the tank and its contents have cooled to 20.0°C, what additional volume of ethanol can be put into the tank?

17.19 •• A machinist bores a hole of diameter 1.30 cm in a steel plate that is at 26.5°C. What is the cross-sectional area of the hole (a) at 26.5°C and (b) when the temperature of the plate is increased to 170°C? Assume that the coefficient of linear expansion remains constant over this temperature range.

17.20 • Consider a flat metal plate with width w and length l, so its area is $A = lw$. The metal has coefficient of linear expansion α. Derive an expression, in terms of α, that gives the change ΔA in area for a change ΔT in temperature.

17.21 •• Steel train rails are laid in 11.0-m-long segments placed end to end. The rails are laid on a winter day when their temperature is −2.5°C. (a) How much space must be left between adjacent rails if they are just to touch on a summer day when their temperature is 35.0°C? (b) If the rails are originally laid in contact, what is the stress in them on a summer day when their temperature is 35.0°C?

17.22 •• A brass rod is 185 cm long and 1.60 cm in diameter. What force must be applied to each end of the rod to prevent it from contracting when it is cooled from 120.0°C to 10.0°C?

17.23 • The increase in length of an aluminum rod is twice the increase in length of an Invar rod with only a third of the temperature increase. Find the ratio of the lengths of the two rods.

Section 17.4 Quantity of Heat

17.24 • Trying to stay awake for an all-night study session, a student makes a cup of coffee by first placing a 200 W electric immersion heater in 0.400 kg of water. (a) How much heat must be added to the water to raise its temperature from 21.0°C to 90.5°C? (b) How much time is required? Assume that all of the heater's power goes into heating the water.

17.25 •• An aluminum tea kettle with mass 1.45 kg and containing 1.90 kg of water is placed on a stove. If no heat is lost to the surroundings, how much heat must be added to raise the temperature from 22.0°C to 84.0°C?

17.26 • **BIO** **Heat Loss During Breathing.** In very cold weather a significant mechanism for heat loss by the human body is energy expended in warming the air taken into the lungs with each breath. (a) On a cold winter day when the temperature is −20°C, what amount of heat is needed to warm to body temperature (37°C) the 0.50 L of air exchanged with each breath? Assume that the specific heat of air is 1020 J/kg · K and that 1.0 L of air has mass 1.3×10^{-3} kg. (b) How much heat is lost per hour if the respiration rate is 20 breaths per minute?

17.27 • **BIO** While running, a 70 kg student generates thermal energy at a rate of 1200 W. For the runner to maintain a constant body temperature of 37°C, this energy must be removed by perspiration or other mechanisms. If these mechanisms failed and the energy could not flow out of the student's body, for what amount of time could a student run before irreversible body damage occurred? (*Note:* Protein structures in the body are irreversibly damaged if body temperature rises to 44°C or higher. The specific heat of a typical human body is 3480 J/kg · K, slightly less than that of water. The difference is due to the presence of protein, fat, and minerals, which have lower specific heats.)

17.28 •• **On-Demand Water Heaters.** Conventional hot-water heaters consist of a tank of water maintained at a fixed temperature. The hot water is to be used when needed. The drawbacks are that energy is wasted because the tank loses heat when it is not in use and that you can run out of hot water if you use too much. Some utility companies are encouraging the use of *on-demand* water heaters (also known as *flash heaters*), which consist of heating units to heat the water as you use it. No water tank is involved, so no heat is wasted. A typical household shower flow rate is 9.0 L/min with the tap water being heated from 10°C to 45°C by the on-demand heater. What rate of heat input (either electrical or from gas) is required to operate such a unit, assuming that all the heat goes into the water?

17.29 • You are given a sample of metal and asked to determine its specific heat. You weigh the sample and find that its weight is 28.4 N. You carefully add 1.25×10^4 J of heat energy to the sample and find that its temperature rises 18.0°C. What is the sample's specific heat?

17.30 • **CP** A 25,000 kg underground train initially traveling at 15.5 m/s slows to a stop at a station and then stays there long enough for its brakes to cool. The station's dimensions are 65.0 m long by 20.0 m wide by 12.0 m high. Assuming all the work done by the brakes in stopping the train is transferred as heat uniformly to all the air in the station, by how much does the air temperature in the station rise? Take the density of the air to be 1.20 kg/m^3 and its specific heat to be 1020 J/kg · K.

17.31 • **CP** While painting the top of an antenna 217 m in height, a worker accidentally lets a 1.00 L water bottle fall from his lunchbox. The bottle lands in some bushes at ground level and does not break. If a quantity of heat equal to the magnitude of the change in mechanical energy of the water goes into the water, what is its increase in temperature?

17.32 • **CP** A nail driven into a board increases in temperature. If we assume that 60% of the kinetic energy delivered by a 1.80 kg hammer with a speed of 7.80 m/s is transformed into heat that flows into the nail and does not flow out, what is the temperature increase of an 8.00 g aluminum nail after it is struck ten times?

17.33 •• **CP** A 15.9 g bullet traveling horizontally at 865 m/s passes through a tank containing 13.5 kg of water and emerges with a speed of 535 m/s. What is the maximum temperature increase that the water could have as a result of this event?

Section 17.5 Calorimetry and Phase Changes

17.34 • You have 750 g of water at 10.0°C in a large insulated beaker. How much boiling water at 100.0°C must you add to this beaker so that the final temperature of the mixture will be 75°C?

17.35 •• A 460.0 g chunk of an unknown metal, which has been in boiling water for several minutes, is quickly dropped into an insulating Styrofoam beaker containing 1.50 kg of water at room temperature (20.0°C). After waiting and gently stirring for 5.00 minutes, you observe that the water's temperature has reached a constant value of 22.0°C. (a) Assuming that the Styrofoam absorbs a negligibly small amount of heat and that no heat was lost to the surroundings, what is the specific heat of the metal? (b) Which is more useful for storing thermal energy: this metal or an equal weight of water? Explain. (c) If the heat absorbed by the Styrofoam actually is not negligible, how would the specific heat you calculated in part (a) be in error? Would it be too large, too small, or still correct? Explain.

17.36 • **BIO** **Treatment for a Stroke.** One suggested treatment for a person who has suffered a stroke is immersion in an ice-water bath at 0°C to lower the body temperature, which prevents damage to the brain. In one set of tests, patients were cooled until their internal temperature reached 32.0°C. To treat a 70.0 kg patient, what is the minimum amount of ice (at 0°C) you need in the bath so that its temperature remains at 0°C? The specific heat of the human body is 3480 J/kg · °C, and recall that normal body temperature is 37.0°C.

17.37 •• A blacksmith cools a 1.20 kg chunk of iron, initially at 650.0°C, by trickling 15.0°C water over it. All of the water boils away, and the iron ends up at 120.0°C. How much water did the blacksmith trickle over the iron?

17.38 •• A copper calorimeter can with mass 0.100 kg contains 0.160 kg of water and 0.0180 kg of ice in thermal equilibrium at atmospheric pressure. If 0.750 kg of lead at 255°C is dropped into the calorimeter can, what is the final temperature? Assume that no heat is lost to the surroundings.

17.39 •• A copper pot with a mass of 0.500 kg contains 0.170 kg of water, and both are at 20.0°C. A 0.250 kg block of iron at 85.0°C is dropped into the pot. Find the final temperature of the system, assuming no heat loss to the surroundings.

17.40 • In a container of negligible mass, 0.200 kg of ice at an initial temperature of −40.0°C is mixed with a mass m of water that has an initial temperature of 80.0°C. No heat is lost to the surroundings. If the final temperature of the system is 28.0°C, what is the mass m of the water that was initially at 80.0°C?

17.41 • A 6.30 kg piece of solid copper metal at an initial temperature T is placed with 2.00 kg of ice that is initially at −27.0°C. The ice is in an insulated container of negligible mass and no heat is exchanged with the surroundings. After thermal equilibrium is reached, there is 0.80 kg of ice and 1.20 kg of liquid water. What was the initial temperature of the piece of copper?

17.42 •• An ice-cube tray of negligible mass contains 0.290 kg of water at 18.0°C. How much heat must be removed to cool the water to 0.00°C and freeze it? Express your answer in joules and calories.

17.43 • How much heat is required to convert 18.0 g of ice at −10.0°C to steam at 100.0°C? Express your answer in joules and calories.

17.44 •• An open container holds 0.550 kg of ice at −15.0°C. The mass of the container can be ignored. Heat is supplied to the container at the constant rate of 800.0 J/min for 500.0 min. (a) After how many minutes does the ice *start* to melt? (b) After how many minutes, from the time when the heating is first started, does the temperature begin to rise above 0.0°C? (c) Plot a curve showing the temperature as a function of the elapsed time.

17.45 • **CP** What must the initial speed of a lead bullet be at 21.0°C so that the heat developed when it is brought to rest will be just sufficient to melt it? Assume that all the initial mechanical energy of the bullet is converted to heat and that no heat flows from the bullet to its

surroundings. (Typical rifles have muzzle speeds that exceed the speed of sound in air, which is 345 m/s at 21.0°C.)

17.46 •• **BIO** **Steam Burns Versus Water Burns.** What is the amount of heat input to your skin when it receives the heat released (a) by 22.0 g of steam initially at 100.0°C, when it is cooled to skin temperature (34.0°C)? (b) By 22.0 g of water initially at 100.0°C, when it is cooled to 34.0°C? (c) What does this tell you about the relative severity of burns from steam versus burns from hot water?

17.47 • **BIO** **"The Ship of the Desert."** Camels require very little water because they are able to tolerate relatively large changes in their body temperature. While humans keep their body temperatures constant to within one or two degrees Celsius, a dehydrated camel permits its body temperature to drop to 34.0°C overnight and rise to 40.0°C during the day. To see how effective this mechanism is for saving water, calculate how many liters of water a 400 kg camel would have to drink if it attempted to keep its body temperature at a constant 34.0°C by evaporation of sweat during the day (12 hours) instead of letting it rise to 40.0°C. (*Note:* The specific heat of a camel or other mammal is about the same as that of a typical human, 3480 J/kg · K. The heat of vaporization of water at 34°C is 2.42×10^6 J/kg.)

17.48 • **BIO** Evaporation of sweat is an important mechanism for temperature control in some warm-blooded animals. (a) What mass of water must evaporate from the skin of a 70.0 kg man to cool his body 1.00°C? The heat of vaporization of water at body temperature (37°C) is 2.42×10^6 J/kg. The specific heat of a typical human body is 3480 J/kg · K (see Exercise 17.27). (b) What volume of water must the man drink to replenish the evaporated water? Compare to the volume of a soft-drink can ($355\ \text{cm}^3$).

17.49 •• **CP** An asteroid with a diameter of 10 km and a mass of 2.60×10^{15} kg impacts the earth at a speed of 32.0 km/s, landing in the Pacific Ocean. If 1.00% of the asteroid's kinetic energy goes to boiling the ocean water (assume an initial water temperature of 10.0°C), what mass of water will be boiled away by the collision? (For comparison, the mass of water contained in Lake Victoria, the largest lake in Africa, is about 2.76×10^{15} kg.)

17.50 • A laboratory technician drops a 0.0850 kg sample of unknown solid material, at 100.0°C, into a calorimeter. The calorimeter can, initially at 19.0°C, is made of 0.150 kg of copper and contains 0.200 kg of water. The final temperature of the calorimeter can and contents is 26.1°C. Compute the specific heat of the sample.

17.51 •• An insulated beaker with negligible mass contains 0.250 kg of water at 73.5°C. How many kilograms of ice at −20.8°C must be dropped into the water to make the final temperature of the system 36.3°C?

17.52 • A 7.00 kg silver ingot is taken from a furnace, where its temperature is 780.0°C, and placed on a large block of ice at 0.0°C. Assuming that all the heat given up by the silver is used to melt the ice, how much ice is melted?

17.53 •• A plastic cup of negligible mass contains 0.280 kg of an unknown liquid at a temperature of 30.0°C. A 0.0270 kg mass of ice at a temperature of 0.0°C is added to the liquid, and when thermal equilibrium is reached the temperature of the combined substances is 14.0°C. Assuming no heat is exchanged with the surroundings, what is the specific heat capacity of the unknown liquid?

Section 17.6 Mechanisms of Heat Transfer

17.54 •• Two rods, one made of brass and the other made of copper, are joined end to end. The length of the brass section is 0.290 m and the length of the copper section is 0.710 m. Each segment has cross-sectional area $6.40 \times 10^{-3}\ \text{m}^2$. The free end of the brass segment is in boiling water and the free end of the copper segment is in an ice–water mixture, in both cases under normal atmospheric pressure. The sides of the rods are insulated so there is no heat loss to the surroundings. (a) What is the temperature of the point where the brass and copper segments are joined? (b) What mass of ice is melted in 8.90 min by the heat conducted by the composite rod?

17.55 •• A copper bar is welded end to end to a bar of an unknown metal. The two bars have the same lengths and cross-sectional areas. The free end of the copper bar is maintained at a temperature T_H that can be varied. The free end of the unknown metal is kept at 0.0°C. To measure the thermal conductivity of the unknown metal, you measure the temperature T at the junction between the two bars for several values of T_H. You plot your data as T versus T_H, both in kelvins, and find that your data are well fit by a straight line that has slope 0.710. What do your measurements give for the value of the thermal conductivity of the unknown metal?

17.56 •• One end of an insulated metal rod is maintained at 100.0°C, and the other end is maintained at 0.00°C by an ice–water mixture. The rod is 70.0 cm long and has a cross-sectional area of $1.15\ \text{cm}^2$. The heat conducted by the rod melts 9.30 g of ice in 15.0 min. Find the thermal conductivity k of the metal.

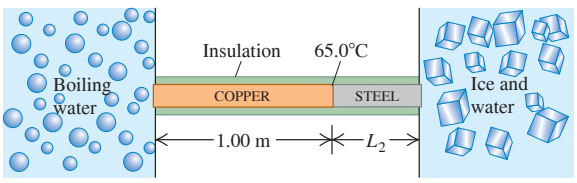
17.57 •• A carpenter builds an exterior house wall with a layer of wood 3.0 cm thick on the outside and a layer of Styrofoam insulation 2.2 cm thick on the inside wall surface. The wood has $k = 0.080\ \text{W/m} \cdot \text{K}$, and the Styrofoam has $k = 0.027\ \text{W/m} \cdot \text{K}$. The interior surface temperature is 19.0°C, and the exterior surface temperature is −10.0°C. (a) What is the temperature at the plane where the wood meets the Styrofoam? (b) What is the rate of heat flow per square meter through this wall?

17.58 • An electric kitchen range has a total wall area of $1.40\ \text{m}^2$ and is insulated with a layer of fiberglass 4.00 cm thick. The inside surface of the fiberglass has a temperature of 175°C, and its outside surface is at 35.0°C. The fiberglass has a thermal conductivity of 0.040 W/m · K. (a) What is the heat current through the insulation, assuming it may be treated as a flat slab with an area of $1.40\ \text{m}^2$? (b) What electric-power input to the heating element is required to maintain this temperature?

17.59 • Air has a very low thermal conductivity. This explains why we feel comfortable wearing short sleeves in a 20°C environment even though our body temperature is 37°C. Material objects feel cooler to our immediate touch than the air, owing to relatively high thermal conductivities. (a) Touch a few surfaces in a room-temperature environment and rank them in order of which feel the coolest to which feel the warmest. Objects that feel cooler have larger thermal conductivities. Consider a wood surface, a metallic surface, and a glass surface, and rank these in order from coolest to warmest. (b) How does your ranking compare to the thermal conductivities listed in Table 17.5?

17.60 • A long rod, insulated to prevent heat loss along its sides, is in perfect thermal contact with boiling water (at atmospheric pressure) at one end and with an ice–water mixture at the other (**Fig. E17.60**). The rod consists of a 1.00 m section of copper (one end in boiling water) joined end to end to a length L_2 of steel (one end in the ice–water mixture). Both sections of the rod have cross-sectional areas of $4.00\ \text{cm}^2$. The temperature of the copper–steel junction is 65.0°C after a steady state has been set up. (a) How much heat per second flows from the boiling water to the ice–water mixture? (b) What is the length L_2 of the steel section?

Figure **E17.60**



17.61 • A pot with a steel bottom 9.00 mm thick rests on a hot stove. The area of the bottom of the pot is 0.130 m^2. The water inside the pot is at 100.0°C, and 0.400 kg are evaporated every 3.00 min. Find the temperature of the lower surface of the pot, which is in contact with the stove.

17.62 •• You are asked to design a cylindrical steel rod 45.0 cm long, with a circular cross section, that will conduct 180.0 J/s from a furnace at 400.0°C to a container of boiling water under 1 atmosphere. What must the rod's diameter be?

17.63 •• A picture window has dimensions of 1.40 m × 2.50 m and is made of glass 6.00 mm thick. On a winter day, the temperature of the outside surface of the glass is −17.0°C, while the temperature of the inside surface is a comfortable 21.0°C. (a) At what rate is heat being lost through the window by conduction? (b) At what rate would heat be lost through the window if you covered it with a 0.750-mm-thick layer of paper (thermal conductivity 0.0500 W/m · K)?

17.64 • What is the rate of energy radiation per unit area of a blackbody at (a) 261 K and (b) 2610 K?

17.65 • **Size of a Light-Bulb Filament.** The operating temperature of a tungsten filament in an incandescent light bulb is 2500 K, and its emissivity is 0.350. Find the surface area of the filament of a 200-W bulb if all the electrical energy consumed by the bulb is radiated by the filament as electromagnetic waves. (Only a fraction of the radiation appears as visible light.)

17.66 •• The emissivity of tungsten is 0.350. A tungsten sphere with radius 1.89 cm is suspended within a large evacuated enclosure whose walls are at 300.0 K. What power input is required to maintain the sphere at 3100.0 K if heat conduction along the supports is ignored?

17.67 • **The Sizes of Stars.** The hot glowing surfaces of stars emit energy in the form of electromagnetic radiation. It is a good approximation to assume $e = 1$ for these surfaces. Find the radii of the following stars (assumed to be spherical): (a) Rigel, the bright blue star in the constellation Orion, which radiates energy at a rate of 2.7×10^{32} W and has surface temperature 11,000 K; (b) Procyon B (visible only using a telescope), which radiates energy at a rate of 2.1×10^{23} W and has surface temperature 10,000 K. (c) Compare your answers to the radius of the earth, the radius of the sun, and the distance between the earth and the sun. (Rigel is an example of a *supergiant* star, and Procyon B is an example of a *white dwarf* star.)

PROBLEMS

17.68 •• Figure 17.12 shows that the graph of the volume of 1 gram of liquid water can be closely approximated by a parabola in the temperature range between 0°C and 10°C. (a) Show that the equation of this parabola has the form $V = A + B(T_C - 4.0°C)^2$ and find the values of the constants A and B. (b) Define the temperature-dependent quantity $\beta(T_C)$ in terms of the equation $dV = \beta(T_C)\, V\, dT$. Use the result of part (a) to find the value of $\beta(T_C)$ for $T_C = 1.0°C$, 4.0°C, 7.0°C, and 10.0°C. Your results show that β is not constant in this temperature range but is approximately constant above 7.0°C.

17.69 •• **CP** A Foucault pendulum consists of a brass sphere with a diameter of 35.0 cm suspended from a steel cable 10.5 m long (both measurements made at 20.0°C). Due to a design oversight, the swinging sphere clears the floor by a distance of only 2.00 mm when the temperature is 20.0°C. At what temperature will the sphere begin to brush the floor?

17.70 •• A steel wire has density 7800 kg/m^3 and mass 2.50 g. It is stretched between two rigid supports separated by 0.400 m. (a) When the temperature of the wire is 20.0°C, the frequency of the fundamental standing wave for the wire is 440 Hz. What is the tension in the wire? (b) What is the temperature of the wire if its fundamental standing wave has frequency 460 Hz? For steel the coefficient of linear expansion is $1.2 \times 10^{-5}\ K^{-1}$ and Young's modulus is 20×10^{10} Pa.

17.71 •• An unknown liquid has density ρ and coefficient of volume expansion β. A quantity of heat Q is added to a volume V of the liquid, and the volume of the liquid increases by an amount ΔV. There is no phase change. In terms of these quantities, what is the specific heat capacity c of the liquid?

17.72 •• **CP** A small fused quartz sphere swings back and forth as a simple pendulum on the lower end of a long copper wire that is attached to the ceiling at its upper end. The amplitude of swing is small. When the wire has a temperature of 20.0°C, its length is 3.00 m. What is the percentage change in the period of the motion if the temperature of the wire is increased to 220°C? (*Hint:* Use the power series expansion for $(1 + x)^n$ in Appendix D.)

17.73 ••• You propose a new temperature scale with temperatures given in °M. You define 0.0°M to be the normal melting point of mercury and 100.0°M to be the normal boiling point of mercury. (a) What is the normal boiling point of water in °M? (b) A temperature change of 19.0°M corresponds to how many °C?

17.74 • **CP CALC** A 250 kg weight is hanging from the ceiling by a thin copper wire. In its fundamental mode, this wire vibrates at the frequency of concert A (440 Hz). You then increase the temperature of the wire by 40°C. (a) By how much will the fundamental frequency change? Will it increase or decrease? (b) By what percentage will the speed of a wave on the wire change? (c) By what percentage will the wavelength of the fundamental standing wave change? Will it increase or decrease?

17.75 ••• You are making pesto for your pasta and have a cylindrical measuring cup 10.0 cm high made of ordinary glass $[\beta = 2.7 \times 10^{-5}\ (°C)^{-1}]$ that is filled with olive oil $[\beta = 6.8 \times 10^{-4}\ (°C)^{-1}]$ to a height of 3.00 mm below the top of the cup. Initially, the cup and oil are at room temperature (22.0°C). You get a phone call and forget about the olive oil, which you inadvertently leave on the hot stove. The cup and oil heat up slowly and have a common temperature. At what temperature will the olive oil start to spill out of the cup?

17.76 •• A surveyor's 30.0 m steel tape is correct at 20.0°C. The distance between two points, as measured by this tape on a day when its temperature is 5.00°C, is 25.970 m. What is the true distance between the points?

17.77 •• A metal rod that is 30.0 cm long expands by 0.0650 cm when its temperature is raised from 0.0°C to 100.0°C. A rod of a different metal and of the same length expands by 0.0350 cm for the same rise in temperature. A third rod, also 30.0 cm long, is made up of pieces of each of the above metals placed end to end and expands 0.0580 cm between 0.0°C and 100.0°C. Find the length of each portion of the composite rod.

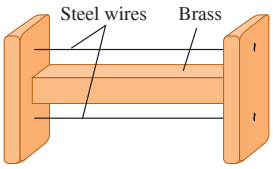
17.78 •• A copper sphere with density 8900 kg/m^3, radius 5.00 cm, and emissivity $e = 1.00$ sits on an insulated stand. The initial temperature of the sphere is 300 K. The surroundings are very cold, so the rate of absorption of heat by the sphere can be neglected. (a) How long does it take the sphere to cool by 1.00 K due to its radiation of heat energy? Neglect the change in heat current as the temperature decreases. (b) To assess the accuracy of the approximation used in part (a), what is the fractional change in the heat current H when the temperature changes from 300 K to 299 K?

17.79 ••• (a) Equation (17.10) gives the stress required to keep the length of a rod constant as its temperature changes. Show that if the length is permitted to change by an amount ΔL when its temperature changes by ΔT, the stress is equal to

$$\frac{F}{A} = Y\left(\frac{\Delta L}{L_0} - \alpha\,\Delta T\right)$$

where F is the tension on the rod, L_0 is the original length of the rod, A its cross-sectional area, α its coefficient of linear expansion, and Y its Young's modulus. (b) A heavy brass bar has projections at its ends (**Fig. P17.79**). Two fine steel wires, fastened between the projections, are just taut (zero tension) when the whole system is at 28°C. What is the tensile stress in the steel wires when the temperature of the system is raised to 130°C? Make any simplifying assumptions you think are justified, but state them.

Figure P17.79



17.80 •• CP A metal wire, with density ρ and Young's modulus Y, is stretched between rigid supports. At temperature T, the speed of a transverse wave is found to be v_1. When the temperature is increased to $T + \Delta T$, the speed decreases to $v_2 < v_1$. Determine the coefficient of linear expansion of the wire.

17.81 •• A steel ring with a 6.2500 cm inside diameter at 20.0°C is to be warmed and slipped over a brass shaft with a 6.2550 cm outside diameter at 20.0°C. (a) To what temperature should the ring be warmed? (b) If the ring and the shaft together are cooled by some means such as liquid air, at what temperature will the ring just slip off the shaft?

17.82 • BIO **Doughnuts: Breakfast of Champions!** A typical doughnut contains 2.0 g of protein, 17.0 g of carbohydrates, and 7.0 g of fat. Average food energy values are 4.0 kcal/g for protein and carbohydrates and 9.0 kcal/g for fat. (a) During heavy exercise, an average person uses energy at a rate of 510 kcal/h. How long would you have to exercise to "work off" one doughnut? (b) If the energy in the doughnut could somehow be converted into the kinetic energy of your body as a whole, how fast could you move after eating the doughnut? Take your mass to be 60 kg, and express your answer in m/s and in km/h.

17.83 •• There is 0.050 kg of an unknown liquid in a plastic container of negligible mass. The liquid has a temperature of 90.0°C. To measure the specific heat capacity of the unknown liquid, you add a mass m_w of water that has a temperature of 0.0°C to the liquid and measure the final temperature T after the system has reached thermal equilibrium. You repeat this measurement for several values of m_w, with the initial temperature of the unknown liquid always equal to 90.0°C. The plastic container is insulated, so no heat is exchanged with the surroundings. You plot your data as m_w versus T^{-1}, the inverse of the final temperature T. Your data points lie close to a straight line that has slope 2.15 kg · °C. What does this result give for the value of the specific heat capacity of the unknown liquid?

17.84 •• You cool a 100.0 g slug of red-hot iron (temperature 745°C) by dropping it into an insulated cup of negligible mass containing 65.0 g of water at 20.0°C. Assuming no heat exchange with the surroundings, (a) what is the final temperature of the water and (b) what is the final mass of the iron and the remaining water?

17.85 •• CALC **Debye's T^3 Law.** At very low temperatures the molar heat capacity of rock salt varies with temperature according to Debye's T^3 law:

$$C = k\frac{T^3}{\theta^3}$$

where $k = 1940$ J/mol · K and $\theta = 281$ K. (a) How much heat is required to raise the temperature of 1.50 mol of rock salt from 13.0 K to 36.0 K? (*Hint:* Use Eq. (17.16) in the form $dQ = nC\,dT$ and integrate.) (b) What is the average molar heat capacity in this range? (c) What is the true molar heat capacity at 36.0 K?

17.86 •• The heat one feels when sitting near the fire in a fireplace or at a campfire is due almost entirely to thermal radiation. (a) Estimate the diameter and length of an average campfire log. (b) Compute the surface area of such a log. (c) Use the Stefan–Boltzmann law to determine the power emitted by thermal radiation by such a log when it burns at a typical temperature of 700°C in a surrounding air temperature of 20.0°C. The emissivity of a burning log is close to unity.

17.87 • **Hot Air in a Physics Lecture.** (a) A typical student listening attentively to a physics lecture has a heat output of 110 W. How much heat energy does a class of 85 physics students release into a lecture hall over the course of a 50 min lecture? (b) Assume that all the heat energy in part (a) is transferred to the 4000 m^3 of air in the room. The air has specific heat 1020 J/kg · K and density 1.20 kg/m^3. If none of the heat escapes and the air conditioning system is off, how much will the temperature of the air in the room rise during the 50 min lecture? (c) If the class is taking an exam, the heat output per student rises to 300 W. What is the temperature rise during 50 min in this case?

17.88 ••• CALC The molar heat capacity of a certain substance varies with temperature according to the empirical equation

$$C = 29.5\ \text{J/mol}\cdot\text{K} + (8.20 \times 10^{-3}\ \text{J/mol}\cdot\text{K}^2)T$$

How much heat is necessary to change the temperature of 3.00 mol of this substance from 27°C to 227°C? (*Hint:* Use Eq. (17.16) in the form $dQ = nC\,dT$ and integrate.)

17.89 •• BIO **Bicycling on a Warm Day.** If the air temperature is the same as the temperature of your skin (about 30°C), your body cannot get rid of heat by transferring it to the air. In that case, it gets rid of the heat by evaporating water (sweat). During bicycling, a typical 70 kg person's body produces energy at a rate of about 500 W due to metabolism, 80% of which is converted to heat. (a) How many kilograms of water must the person's body evaporate in an hour to get rid of this heat? The heat of vaporization of water at body temperature is 2.42×10^6 J/kg. (b) The evaporated water must, of course, be replenished, or the person will dehydrate. How many 750 mL bottles of water must the cyclist drink per hour to replenish the lost water? (Recall that the mass of a liter of water is 1.0 kg.)

17.90 •• BIO **Overheating.** (a) By how much would the body temperature of the cyclist in Problem 17.89 increase in an hour if he were unable to get rid of the excess heat? (b) Is this temperature increase large enough to be serious? To find out, how high a fever would it be equivalent to? (Recall that the normal internal body temperature is 37.0°C and the specific heat of the body is 3480 J/kg · °C.)

17.91 • BIO **A Thermodynamic Process in an Insect.** The African bombardier beetle (*Stenaptinus insignis*) can emit a jet of defensive spray from the movable tip of its abdomen (**Fig. P17.91**). The beetle's body has reservoirs containing two chemicals; when the beetle is disturbed, these chemicals combine in a reaction chamber, producing a compound that is warmed from 20°C to 100°C by the heat of reaction. The high pressure produced allows the compound to be sprayed out at speeds up to 19 m/s (68 km/h), scaring away predators of all kinds. (The beetle shown in Fig. P17.91 is 2 cm long.) Calculate the heat of reaction of the two chemicals (in J/kg). Assume that the specific heat of the chemicals and of the spray is the same as that of water, 4.19×10^3 J/kg · K, and that the initial temperature of the chemicals is 20°C.

Figure P17.91

17.92 ••• CP An industrious explorer of the polar regions has devised a contraption for melting ice. It consists of a sealed 10 L cylindrical tank with a porous grate separating the top half from the bottom half. The bottom half includes a paddle wheel attached to an axle that passes outside the cylinder, where it is attached by a gearbox and pulley system to a stationary bicycle. Pedaling the bicycle rotates the paddle wheel inside the cylinder. The tank includes 6.00 L of water and 3.00 kg of ice at 0.0°C. The water fills the bottom chamber, where it may be agitated by the paddle wheel, and partially fills the upper chamber, which also includes the ice. The bicycle is pedaled with an average torque of 25.0 N · m at a rate of 30.0 revolutions per minute. The system is 70% efficient. (a) For what length of time must the explorer pedal the bicycle to melt all the ice? (b) How much longer must he pedal to raise the temperature of the water to 10.5°C?

17.93 •• You have 1.50 kg of water at 28.0°C in an insulated container of negligible mass. You add 0.600 kg of ice that is initially at −22.0°C. Assume that no heat exchanges with the surroundings. (a) After thermal equilibrium has been reached, has all of the ice melted? (b) If all of the ice has melted, what is the final temperature of the water in the container? If some ice remains, what is the final temperature of the water in the container, and how much ice remains?

17.94 •• A thirsty nurse cools a 2.00 L bottle of a soft drink (mostly water) by pouring it into a large aluminum mug of mass 0.257 kg and adding 0.120 kg of ice initially at −15.0°C. If the soft drink and mug are initially at 20.0°C, what is the final temperature of the system, assuming that no heat is lost?

17.95 ••• A copper calorimeter can with mass 0.450 kg contains 0.0940 kg of ice. The system is initially at 0.0°C. (a) If 0.0350 kg of steam at 100.0°C and 1.00 atm pressure is added to the can, what is the final temperature of the calorimeter can and its contents? (b) At the final temperature, how many kilograms are there of ice, how many of liquid water, and how many of steam?

17.96 • A Styrofoam bucket of negligible mass contains 1.75 kg of water and 0.450 kg of ice. More ice, from a refrigerator at −15.0°C, is added to the mixture in the bucket, and when thermal equilibrium has been reached, the total mass of ice in the bucket is 0.884 kg. Assuming no heat exchange with the surroundings, what mass of ice was added?

17.97 ••• In a container of negligible mass, 0.0400 kg of steam at 100°C and atmospheric pressure is added to 0.200 kg of water at 50.0°C. (a) If no heat is lost to the surroundings, what is the final temperature of the system? (b) At the final temperature, how many kilograms are there of steam and how many of liquid water?

17.98 •• BIO **Mammal Insulation.** Animals in cold climates often depend on *two* layers of insulation: a layer of body fat (of thermal conductivity 0.20 W/m · K) surrounded by a layer of air trapped inside fur or down. We can model a black bear (*Ursus americanus*) as a sphere 1.5 m in diameter having a layer of fat 3.90 cm thick. (Actually, the thickness varies with the season, but we are interested in hibernation, when the fat layer is thickest.) In studies of bear hibernation, it was found that the outer surface layer of the fur is at 2.90°C during hibernation, while the inner surface of the fat layer is at 31.3°C. (a) What is the temperature at the fat–inner fur boundary so that the bear loses heat at a rate of 50.8 W? (b) How thick should the air layer (contained within the fur) be?

17.99 •• **Effect of a Window in a Door.** A carpenter builds a solid wood door with dimensions 2.10 m × 0.91 m × 6.0 cm. Its thermal conductivity is $k = 0.120$ W/m · K. The air films on the inner and outer surfaces of the door have the same combined thermal resistance as an additional 1.9 cm thickness of solid wood. The inside air temperature is 21.0°C, and the outside air temperature is −5.0°C. (a) What is the rate of heat flow through the door? (b) By what factor is the heat flow increased if a window 0.540 m on a side is inserted in the door? The glass is 0.5 cm thick, and the glass has a thermal conductivity of 0.80 W/m · K. The air films on the two sides of the glass have a total thermal resistance that is the same as an additional 12.0 cm of glass.

17.100 ••• CP At 0°C, a cylindrical metal bar with radius r and mass M is slid snugly into a circular hole in a large, horizontal, rigid slab of thickness d. For this metal, Young's modulus is Y and the coefficient of linear expansion is α. A light but strong hook is attached to the underside of the metal bar; this apparatus is used as part of a hoist in a shipping yard. The coefficient of static friction between the bar and the slab is μ_s. At a temperature T above 0°C, the hook is attached to a large container and the slab is raised. What is the largest mass the container can have without the metal bar slipping out of the slab as the container is slowly lifted? The slab undergoes negligible thermal expansion.

17.101 •• Compute the ratio of the rate of heat loss through a single-pane window with area 0.15 m^2 to that for a double-pane window with the same area. The glass of a single pane is 4.2 mm thick, and the air space between the two panes of the double-pane window is 7.0 mm thick. The glass has thermal conductivity 0.80 W/m · K. The air films on the room and outdoor surfaces of either window have a combined thermal resistance of 0.15 $m^2 \cdot K/W$.

17.102 • Rods of copper, brass, and steel—each with cross-sectional area of 2.00 cm^2—are welded together to form a Y-shaped figure. The free end of the copper rod is maintained at 100.0°C, and the free ends of the brass and steel rods at 0.0°C. Assume that there is no heat loss from the surfaces of the rods. The lengths of the rods are: copper, 13.0 cm; brass, 18.0 cm; steel, 24.0 cm. What is (a) the temperature of the junction point; (b) the heat current in each of the three rods?

17.103 •• BIO **Jogging in the Heat of the Day.** You have probably seen people jogging in extremely hot weather. There are good reasons not to do this! When jogging strenuously, an average runner of mass 68 kg and surface area 1.85 m^2 produces energy at a rate of up to 1300 W, 80% of which is converted to heat. The jogger radiates heat but actually absorbs more from the hot air than he radiates away. At such high levels of activity, the skin's temperature can be elevated to around 33°C instead of the usual 30°C. (Ignore conduction, which would bring even more heat into his body.) The only way for the body to get rid of this extra heat is by evaporating water (sweating). (a) How much heat per second is produced just by the act of jogging? (b) How much *net* heat per second does the runner gain just from radiation if the air temperature is 40.0°C? (Remember: He radiates out, but the environment radiates back in.) (c) What is the *total* amount of excess heat this runner's body must get rid of per second? (d) How much water must his body evaporate every minute due to his activity? The heat of vaporization of water at body temperature is 2.42×10^6 J/kg. (e) How many 750 mL bottles of water must he drink after (or preferably before!) jogging for a half hour? Recall that a liter of water has a mass of 1.0 kg.

17.104 ••• BIO **Basal Metabolic Rate.** The *basal metabolic rate* is the rate at which energy is produced in the body when a person is at rest. A 75 kg person of height 1.83 m has a body surface area of approximately 2.0 m^2. (a) What is the net amount of heat this person could radiate per second into a room at 18°C if his skin's surface temperature is 30°C? (At such temperatures, nearly all the heat is infrared radiation, for which the body's emissivity is 1.0, regardless of the amount of pigment.) (b) Normally, 80% of the energy produced by metabolism goes into heat, while the rest goes into things like pumping blood and repairing cells. Also normally, a person at rest can get rid of this excess heat just through radiation. Use your answer to part (a) to find this person's basal metabolic rate.

17.105 ••• CALC **Time Needed for a Lake to Freeze Over.** (a) When the air temperature is below 0°C, the water at the surface of a lake freezes to form an ice sheet. Why doesn't freezing occur throughout the entire volume of the lake? (b) Show that the thickness of the ice sheet formed on the surface of a lake is proportional to the square root of the time if the heat of fusion of the water freezing on the underside of the ice sheet is conducted through the sheet. (c) Assuming that the upper surface of the ice sheet is at −10°C and the bottom surface is at 0°C, calculate the time it will take to form an ice sheet 25 cm thick. (d) If the lake in part (c) is uniformly 40 m deep, how long would it take to freeze all the water in the lake? Is this likely to occur?

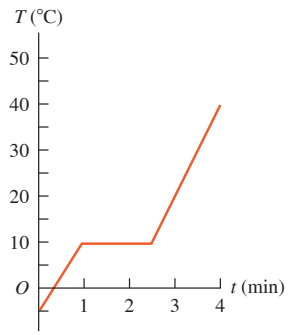
17.106 • The rate at which radiant energy from the sun reaches the earth's upper atmosphere is about $1.50\ \text{kW/m}^2$. The distance from the earth to the sun is 1.50×10^{11} m, and the radius of the sun is 6.96×10^8 m. (a) What is the rate of radiation of energy per unit area from the sun's surface? (b) If the sun radiates as an ideal blackbody, what is the temperature of its surface?

17.107 ••• **A Thermos for Liquid Helium.** A physicist uses a cylindrical metal can 0.250 m high and 0.090 m in diameter to store liquid helium at 4.22 K; at that temperature the heat of vaporization of helium is 2.09×10^4 J/kg. Completely surrounding the metal can are walls maintained at the temperature of liquid nitrogen, 77.3 K, with vacuum between the can and walls. How much liquid helium boils away per hour? The emissivity of the metal can is 0.200. The only heat transfer between the metal can and the surrounding walls is by radiation.

17.108 •• A metal sphere with radius 3.20 cm is suspended in a large metal box with interior walls that are maintained at 30.0°C. A small electric heater is embedded in the sphere. Heat energy must be supplied to the sphere at the rate of 0.660 J/s to maintain the sphere at a constant temperature of 41.0°C. (a) What is the emissivity of the metal sphere? (b) What power input to the sphere is required to maintain it at 82.0°C? What is the ratio of the power required for 82.0°C to the power required for 41.0°C? How does this ratio compare with 2^4? Explain.

17.109 •• DATA As a physicist, you put heat into a 500.0 g solid sample at the rate of 10.0 kJ/min while recording its temperature as a function of time. You plot your data as shown in **Fig. P17.109**. (a) What is the latent heat of fusion for this solid? (b) What are the specific heats of the liquid and solid states of this material?

Figure P17.109



17.110 ••• DATA At a chemical plant where you are an engineer, a tank contains an unknown liquid. You must determine the liquid's specific heat capacity. You put 0.500 kg of the liquid into an insulated metal cup of mass 0.200 kg. Initially the liquid and cup are at 20.0°C. You add 0.500 kg of water that has a temperature of 80.0°C. After thermal equilibrium has been reached, the final temperature of the two liquids and the cup is 58.1°C. You then empty the cup and repeat the experiment with the same initial temperatures, but this time with 1.00 kg of the unknown liquid. The final temperature is 49.3°C. Assume that the specific heat capacities are constant over the temperature range of the experiment and that no heat is lost to the surroundings. Calculate the specific heat capacity of the liquid and of the metal from which the cup is made.

17.111 •• DATA As a mechanical engineer, you are given two uniform metal bars A and B, made from different metals, to determine their thermal conductivities. You measure both bars to have length 40.0 cm and uniform cross-sectional area $2.50\ \text{cm}^2$. You place one end of bar A in thermal contact with a very large vat of boiling water at 100.0°C and the other end in thermal contact with an ice–water mixture at 0.0°C. To prevent heat loss along the bar's sides, you wrap insulation around the bar. You weigh the amount of ice initially as 300 g. After 45.0 min, you weigh the ice again; 191 g of ice remains. The ice–water mixture is in an insulated container, so the only heat entering or leaving it is the heat conducted by the metal bar.

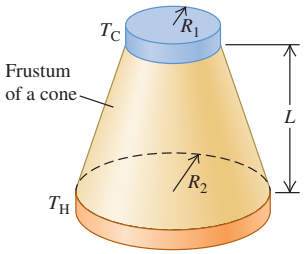
You are confident that your data will allow you to calculate the thermal conductivity k_A of bar A. But this measurement was tedious—you don't want to repeat it for bar B. Instead, you glue the bars together end to end, with adhesive that has very large thermal conductivity, to make a composite bar 80.0 m long. You place the free end of A in thermal contact with the boiling water and the free end of B in thermal contact with the ice–water mixture. The composite bar is thermally insulated. Hours later, you notice that ice remains in the ice–water mixture. Measuring the temperature at the junction of the two bars, you find that it is 62.4°C. After 10 min you repeat that measurement and get the same temperature, with ice remaining in the ice–water mixture. From your data, calculate the thermal conductivities of bar A and of bar B.

CHALLENGE PROBLEMS

17.112 • At a remote arctic research base, liquid water is obtained by melting ice in a propane-fueled conversion tank. Propane has a heat of combustion of 25.6 MJ/L, and 30% of the released energy supplies heat to the tank. Liquid water at 0°C is drawn off the tank at a rate of 500 mL/min, while a corresponding amount of ice at 0°C is continually inserted into the tank from a hopper. How long will an 18 L tank of propane fuel this operation?

17.113 ••• CALC A frustum of a cone (**Fig. P17.113**) has smaller radius R_1, larger radius R_2, and length L and is made from a material with thermal conductivity k. Derive an expression for the conductive heat current through the frustum when the side with radius R_1 is kept at temperature T_H and the side with radius R_2 is kept at temperature T_C. [*Hint:* Parameterize the axis of the frustum using coordinate x. Use Eq. (17.19) for the heat current H through a differential slice of the frustum with length dx, area $A = \pi r^2$ (where r is a function of x), and temperature difference dT. Separate variables and integrate on dT and dx.]

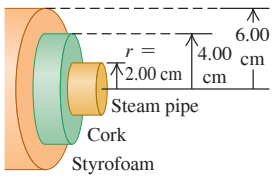
Figure P17.113



17.114 ••• BIO **A Walk in the Sun.** Consider a poor lost soul walking at 5 km/h on a hot day in the desert, wearing only a bathing suit. This person's skin temperature tends to rise due to four mechanisms: (i) energy is generated by metabolic reactions in the body at a rate of 280 W, and almost all of this energy is converted to heat that flows to the skin; (ii) heat is delivered to the skin by convection from the outside air at a rate equal to $k'A_{skin}(T_{air} - T_{skin})$, where k' is $54\ \text{J/h}\cdot\text{°C}\cdot\text{m}^2$, the exposed skin area A_{skin} is $1.5\ \text{m}^2$, the air temperature T_{air} is 47°C, and the skin temperature T_{skin} is 36°C; (iii) the skin absorbs radiant energy from the sun at a rate of $1400\ \text{W/m}^2$; (iv) the skin absorbs radiant energy from the environment, which has temperature 47°C. (a) Calculate the net rate (in watts) at which the person's skin is heated by all four of these mechanisms. Assume that the emissivity of the skin is $e = 1$ and that the skin temperature is initially 36°C. Which mechanism is the most important? (b) At what rate (in L/h) must perspiration evaporate from this person's skin to maintain a constant skin temperature? (The heat of vaporization of water at 36°C is 2.42×10^6 J/kg.) (c) Suppose the person is protected by light-colored clothing ($e \approx 0$) and only $0.45\ \text{m}^2$ of skin is exposed. What rate of perspiration is required now? Discuss the usefulness of the traditional clothing worn by desert peoples.

17.115 ••• A hollow cylinder has length L, inner radius a, and outer radius b, and the temperatures at the inner and outer surfaces are T_2 and T_1. (The cylinder could represent an insulated hot-water pipe.) The thermal conductivity of the material of which the cylinder is made is k. Derive an equation for (a) the total heat current through the walls of the cylinder; (b) the temperature variation inside the cylinder walls. (c) Show that the equation for the total heat current reduces to Eq. (17.19) for linear heat flow when the cylinder wall is very thin. (d) A steam pipe with a radius of 2.00 cm, carrying steam at 140°C, is surrounded by a cylindrical jacket with inner and outer radii 2.00 cm and 4.00 cm and made of a type of cork with thermal conductivity 4.00×10^{-2} W/m · K. This in turn is surrounded by a cylindrical jacket made of a brand of Styrofoam with thermal conductivity 2.70×10^{-2} W/m · K and having inner and outer radii 4.00 cm and 6.00 cm (**Fig. P17.115**). The outer surface of the Styrofoam has a temperature of 15°C. What is the temperature at a radius of 4.00 cm, where the two insulating layers meet? (e) What is the total rate of transfer of heat out of a 2.00 m length of pipe?

Figure P17.115



MCAT-STYLE PASSAGE PROBLEMS

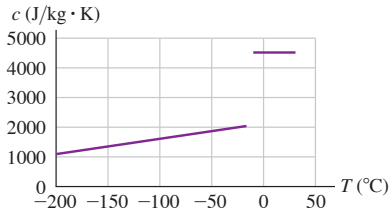
BIO Preserving Cells at Cold Temperatures. In cryopreservation, biological materials are cooled to a very low temperature to slow down chemical reactions that might damage the cells or tissues. It is important to prevent the materials from forming ice crystals during freezing. One method for preventing ice formation is to place the material in a protective solution called a *cryoprotectant*. Stated values of the thermal properties of one cryoprotectant are listed here:

Melting point	−20°C
Latent heat of fusion	2.80×10^5 J/kg
Specific heat (liquid)	4.5×10^3 J/kg · K
Specific heat (solid)	2.0×10^3 J/kg · K
Thermal conductivity (liquid)	1.2 W/m · K
Thermal conductivity (solid)	2.5 W/m · K

17.116 You place 35 g of this cryoprotectant at 22°C in contact with a cold plate that is maintained at the boiling temperature of liquid nitrogen (77 K). The cryoprotectant is thermally insulated from everything but the cold plate. Use the values in the table to determine how much heat will be transferred from the cryoprotectant as it reaches thermal equilibrium with the cold plate. (a) 1.5×10^4 J; (b) 2.9×10^4 J; (c) 3.4×10^4 J; (d) 4.4×10^4 J.

17.117 Careful measurements show that the specific heat of the solid phase depends on temperature (**Fig. P17.117**). How will the actual time needed for this cryoprotectant to come to equilibrium with the cold plate compare with the time predicted by using the values in the table? Assume that all values other than the specific heat (solid) are correct. The actual time (a) will be shorter; (b) will be longer; (c) will be the same; (d) depends on the density of the cryoprotectant.

Figure P17.117



17.118 In another experiment, you place a layer of this cryoprotectant between one 10 cm × 10 cm cold plate maintained at −40°C and a second cold plate of the same size maintained at liquid nitrogen's boiling temperature (77 K). Then you measure the rate of heat transfer. Another lab wants to repeat the experiment but uses cold plates that are 20 cm × 20 cm, with one at −40°C and the other at 77 K. How thick does the layer of cryoprotectant have to be so that the rate of heat transfer by conduction is the same as that when you use the smaller plates? (a) One-quarter the thickness; (b) half the thickness; (c) twice the thickness; (d) four times the thickness.

17.119 To measure the specific heat in the liquid phase of a newly developed cryoprotectant, you place a sample of the new cryoprotectant in contact with a cold plate until the solution's temperature drops from room temperature to its freezing point. Then you measure the heat transferred to the cold plate. If the system isn't sufficiently isolated from its room-temperature surroundings, what will be the effect on the measurement of the specific heat? (a) The measured specific heat will be greater than the actual specific heat; (b) the measured specific heat will be less than the actual specific heat; (c) there will be no effect because the thermal conductivity of the cryoprotectant is so low; (d) there will be no effect on the specific heat, but the temperature of the freezing point will change.

ANSWERS

Chapter Opening Question ?

(iii) The molten iron contains a large amount of energy. An object *has* a temperature but does not *contain* temperature. By "heat" we mean energy that is in transit from one object to another as a result of temperature difference between the objects. Objects do not *contain* heat.

Key Example VARIATION Problems

VP17.4.1 (a) 2.00×10^{-5} K^{-1} (b) length decreases by 0.270 mm

VP17.4.2 12 cm^3

VP17.4.3 1.7×10^3 N; tensile

VP17.4.4 $\dfrac{1}{\alpha_B - \alpha_A}\left(\alpha_B L - \dfrac{\Delta L}{\Delta T}\right)$

VP17.9.1 0.20 kg

VP17.9.2 14.0°C

VP17.9.3 0.181 kg

VP17.9.4 (a) 3.31°C (b) all of it

VP17.15.1 (a) 0.754 W/m · K (b) 733 W

VP17.15.2 (a) 254°C (b) −31°C

VP17.15.3 1.2×10^9 m; 1.7 times the sun's radius

VP17.15.4 4.4×10^4 W

Bridging Problem

(a) 14.26 K (b) 0.427 kg/h